# 100 Python Programs

## 1–25 | Python Fundamentals + Logic

☐ 1. Print text, numbers and formatted output
☐ 2. Take user input and display a personalized message
☐ 3. Add, subtract, multiply and divide two numbers
☐ 4. Calculate area and perimeter of a rectangle
☐ 5. Calculate area and circumference of a circle
☐ 6. Convert Celsius to Fahrenheit and Fahrenheit to Celsius
☐ 7. Calculate simple interest
☐ 8. Calculate compound interest
☐ 9. Swap two numbers — with and without a temporary variable
☐ 10. Check whether a number is positive, negative or zero
☐ 11. Check whether a number is even or odd
☐ 12. Find the largest of two numbers
☐ 13. Find the largest of three numbers
☐ 14. Check whether a year is a leap year
☐ 15. Check whether a character is a vowel or consonant
☐ 16. Build a simple calculator using if/elif
☐ 17. Find the sum of first N natural numbers
☐ 18. Find factorial of a number
☐ 19. Print multiplication table of a number
☐ 20. Print numbers from 1 to N and N to 1
☐ 21. Find sum of digits of a number
☐ 22. Count digits in a number
☐ 23. Reverse a number
☐ 24. Check whether a number is a palindrome
☐ 25. Check whether a number is an Armstrong number

## 26–45 | Loops, Numbers, Patterns & Functions

☐ 26. Check whether a number is prime
☐ 27. Print all prime numbers in a range
☐ 28. Count the number of primes in a range
☐ 29. Find GCD/HCF of two numbers
☐ 30. Find LCM of two numbers
☐ 31. Generate Fibonacci series using a loop
☐ 32. Generate Fibonacci series using a function
☐ 33. Find whether a number belongs to Fibonacci sequence
☐ 34. Calculate x raised to n without using **
☐ 35. Find all factors/divisors of a number
☐ 36. Check whether a number is a perfect number
☐ 37. Check whether a number is a strong number
☐ 38. Convert decimal to binary without bin()
☐ 39. Convert binary to decimal without int(..., 2)
☐ 40. Find the frequency of digits in a number
☐ 41. Print star right triangle
☐ 42. Print inverted star triangle
☐ 43. Print pyramid and inverted pyramid patterns
☐ 44. Print number patterns (1, 12, 123... style)
☐ 45. Write reusable functions for prime, factorial, GCD and palindrome checks

## 46–70 | Strings, Lists, Tuples, Sets & Dictionaries

☐ 46. Reverse a string without slicing
☐ 47. Check whether a string is a palindrome
☐ 48. Count vowels, consonants, digits and spaces in a string
☐ 49. Count frequency of each character
☐ 50. Find the first non-repeating character

☐ 51. Remove duplicate characters while preserving order
☐ 52. Check whether two strings are anagrams
☐ 53. Find the longest word in a sentence
☐ 54. Find the most frequent word in a sentence
☐ 55. Implement string compression (e.g., aaabb → a3b2)
☐ 56. Find the second-largest element in a list
☐ 57. Find the second-smallest element in a list
☐ 58. Remove duplicates from a list without using set()
☐ 59. Find frequency of every element in a list
☐ 60. Find common elements of two lists
☐ 61. Find union and intersection of two lists
☐ 62. Move all zeros to the end of a list
☐ 63. Rotate a list left/right by K positions
☐ 64. Find missing number from 1..N
☐ 65. Find duplicate element(s) in a list
☐ 66. Two Sum: find two elements whose sum equals target
☐ 67. Merge two sorted lists
☐ 68. Find maximum and minimum without max()/min()
☐ 69. Flatten a nested list by one level
☐ 70. Create a word-frequency dictionary from a sentence
☐ 71. Group/list elements by frequency using a dictionary

## 71–85 | Problem-Solving + Searching/Sorting

☐ 72. Linear search
☐ 73. Binary search (iterative)
☐ 74. Binary search (recursive)
☐ 75. Bubble sort
☐ 76. Selection sort
☐ 77. Insertion sort
☐ 78. Merge sort
☐ 79. Quick sort
☐ 80. Count swaps/inversions using a simple approach
☐ 81. Find Kth largest/smallest element
☐ 82. Find the first and last occurrence of a target in a sorted list
☐ 83. Find insertion position of an element in a sorted list
☐ 84. Find a pair with a given difference
☐ 85. Find maximum subarray sum (Kadane's algorithm)
☐ 86. Find longest increasing contiguous subarray

## 86–100 | DSA Readiness: Stack, Queue, Linked List, Recursion & Trees

☐ 87. Implement a stack using a Python list
☐ 88. Implement a queue using collections.deque
☐ 89. Check balanced parentheses using a stack
☐ 90. Evaluate a postfix expression using a stack
☐ 91. Convert infix expression to postfix
☐ 92. Implement a circular queue
☐ 93. Implement a singly linked list: insert at beginning/end
☐ 94. Traverse and search a singly linked list
☐ 95. Delete a node from a singly linked list
☐ 96. Reverse a singly linked list
☐ 97. Find middle node of a linked list
☐ 98. Detect a cycle in a linked list (Floyd's algorithm)
☐ 99. Implement recursion for factorial and Fibonacci
☐ 100. Generate all subsets of a small list using recursion/backtracking
☐ 101. Generate permutations of a string/list
☐ 102. Implement a binary tree node and preorder/inorder/postorder traversal
☐ 103. Level-order traversal using a queue
☐ 104. Find height/depth of a binary tree

☐ 105. Binary Search Tree: insert and search
☐ 106. Binary Search Tree: find minimum/maximum and validate BST